

APM1110 FA6 Cellona, Bryan Dion

Github link: <https://github.com/bryandioncellona/APM1110/blob/main/FA6/APM1110%20FA6%20Cellona%2C%20Bryan%20Dion.Rmd>

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I. Geometric Distribution

Provide an R code for the geometric distribution. The geometric distribution is a probability distribution that models the number of trials required to achieve the first success in a sequence of Bernoulli trials, where each trial has a constant probability of success.

1. Set the probability of success

```
# Each trial has a 20% probability of success.  
p <- 0.2
```

2. Generate 1000 geometric random variables

```
# Fix the random seed and RNG settings so the results can be reproduced.  
RNGkind(kind = "Mersenne-Twister", normal.kind = "Inversion",  
        sample.kind = "Rejection")  
set.seed(1110)  
# rgeom() counts failures; add 1 to include the successful trial.  
x <- rgeom(n = 1000, prob = p) + 1
```

There are 1000 simulated observations. Each observation represents one sequence of trials ending at its first success; it does not represent a single Bernoulli trial.

3. Calculate basic statistics

```
# Calculate the sample mean, sample variance, and sample standard deviation.  
# var() uses the sample denominator n - 1.  
mean_x <- mean(x)  
var_x <- var(x)  
sd_x <- sd(x)
```

4. Print the required results

The first output lists all 1000 simulated trial counts. `sprintf()` formats each summary statistic with exactly two digits after the decimal point.

```
cat("Number of trials required to achieve first success:")
```

```
## Number of trials required to achieve first success:
```

```
print(x)
```

```
##      [1]  4 11  1  1  4  2  9 14 14  3 15  4  9 16  3  7  6  3  7  5  9  7  2  2
##     [25]  1  8 10  7  1  5  4  5  8 34  9 10  1 13  1  1  2  1  4  2  1  2  2  4
##    [49]  2  4  2  1  5  2  2  4  2  1  4  2  2  3  1  5 12  1  8  5  2  6  6  4
##    [73]  6  7  1  1  6  3  1  9  9  1  3 21  3  2 11  2  2  3 16  3  6  1  5
##    [97]  8  3  6  1  4  5  2  3  1  4  9  6  7 19  1  7  1  4  9 10  7  5  1  5
##   [121]  6  1  1  2  2  2  3  3 26  1  5  6  1  6 16  6  6  7  4  7  3  4  3  3
##   [145] 10  2  3 11  2  2  1  1  5  8  2  1  3  2  4 10 10 12 18  3  1  5  1  5
##   [169]  4  3  5  3  1  2 10  2  3 11  1  6  8  1  2 10  3  2  4  1  4 18  1  7
##   [193]  2  9  1  2  2  7  1  1  3 21  7  1  6  4  3  8  4  3  9  1  1 14  4  7
##   [217]  1  1 10  3  2  2  8  3  8  9  4  1  1 31  6 13  2  4  1  5  6  6  4  7
##   [241]  9 11  3  9  4 10 10  1  1  2 12 11  6  4  4  1  3  2  3 17  5 12  5  3
##   [265]  6 11 12  3  8  1  8  2  3  5  1  2  1  4  4  5  3  1  5  5  1 15  1  7
##   [289]  1  3  2  3  2  3 13 11  3  2  9  1  3  8 11  5  2  1  5  2  2  3  4 10
##   [313]  1 21  1  7  1  4  4  8 11  2  3  1  3  1  8  1  8  9  2  1  5  1  2  6
##   [337]  2  7  4  4  2  5  2  1  1 17 13  4  1  9  1  1  3  1 12  4  1  9  6 16
##   [361]  7  1  3  3  1 10  3  1  6  5  5  5 10  3  1 12  9  1  1 13  7  4  6  5
##   [385]  8  2  3 15  1  3  7  4  1  7  5  2  6  4  5  1  8  1  2  4  1  4  1  9
##   [409]  2  3  3  6  6 11 18  8  2  1  3  1  3  2  2  2  1  3  5  2 14  2  7  7
##   [433] 14  1  1  4 15  2  5  3 13  2  1 11  4  2  1  4  2  5  2  1  4  1  4  1
##   [457] 12  3  3  2  3  2  2  1  3  9  1  4 10  3  1  3  7  1  4  7 14 15  4 10
##   [481]  4  2  5 10  6  2  2  9  8  4  1  7  7  2  3  5  2  6 13 19  2  2  2 11
##   [505] 16  9  4  1  2  2  5  2  2 10  6  8 10  8  3 14  6  2  1  9 18  8  4  3
##   [529]  2  1  1 10  3  8 11  8  9  1  2  2 10  2  3  1  1  7  1  3  1  1  1 10
##   [553]  2  1  1  2  7  3  2  3  5  2  2  2  1  1  1  2  3  5  5  8  2  4  2  6
##   [577]  6  5 13  4  2  1  5  1  1 15  2  2  2  1  6 10  3 10  6  3  1  9  2  1
##   [601]  6  3  4  6  5  5  4 14  1  5  8 10  8  4 11  2  3  4  2  3  1  2  8 10
##   [625]  4 10  9  6  6  4  7  4  2  6  8  5  1  1  4 20  3  5  1  5  4 11  3 11
##   [649]  1  2  9  7  6 15  8  6  3  3  4  3  1  2  7  3 30  1  2  1  2  5  2  2
##   [673]  6  7  2  4  4  2  1  9  1  1  3  2 10  3  6  2  8  3  6 22  2  2  5 15
##   [697] 11  2  3 10  3  5  2  7 16  2  4  4  7  5  9  6 12  2 10  5  3  2 11  1
##   [721]  2  3  3  4  1  1  1  1  2  1  3  1 14  2 11  4 11 40 12  4  6  8  2  6
##   [745]  1  2  8 14 11  4  6 11  6  3  5  1  6  1  5  1  2  1  2  3  5 34  1  2
##   [769]  2  2  2  6 10  2  2  3  2  4  9  2  4  7  3  2  6  2  8  1  3  1  8  6
##   [793]  1  8  2  2  5  1  3  5  4  3  2  7  6 14  4  7  2  3  1 17  1  4  5  2
##   [817]  1 14  5  3  1  9  2  1  5  2  6  3  4  5 10  2  2  3  6 11  2  7  1  1
##   [841]  6 12  1  3  8  1  1  2  9  6  3 14 16  9  8  4  1  4  3  1  7  5  5  2
##   [865]  3  3  3  1  3 10  2 18  2  2  1  1  6 10  2  4  3  7  2  5  6  3  5 13
##   [889]  1  2  2  1  3  2  5  3  1  1  4 15  1  2  3  8 18  4  4  2 12  7  1  1
##   [913]  5 10  8  4  7  2  1  1 10  1 12  2  6  4 10  6  7  2  2  1  3  2 13  6
##   [937] 19  1  3 14 11  2  2  1 14  2  6  4  3 21  4  7  8  1  8  2  9  2  1  2
##   [961]  1  6  1 60  1  6  7  4  7  1  5  1  5  3 10  3 12  4  3 11  1  2  1  1
##   [985]  4  1  1  6 26  8  1  3  1  7  3  3  1  1  6  1
```

```
cat(sprintf("Mean (in 2 decimal places): %.2f", mean_x))
```

```
## Mean (in 2 decimal places): 5.09
```

```
cat(sprintf("Variance (in 2 decimal places): %.2f", var_x))
```

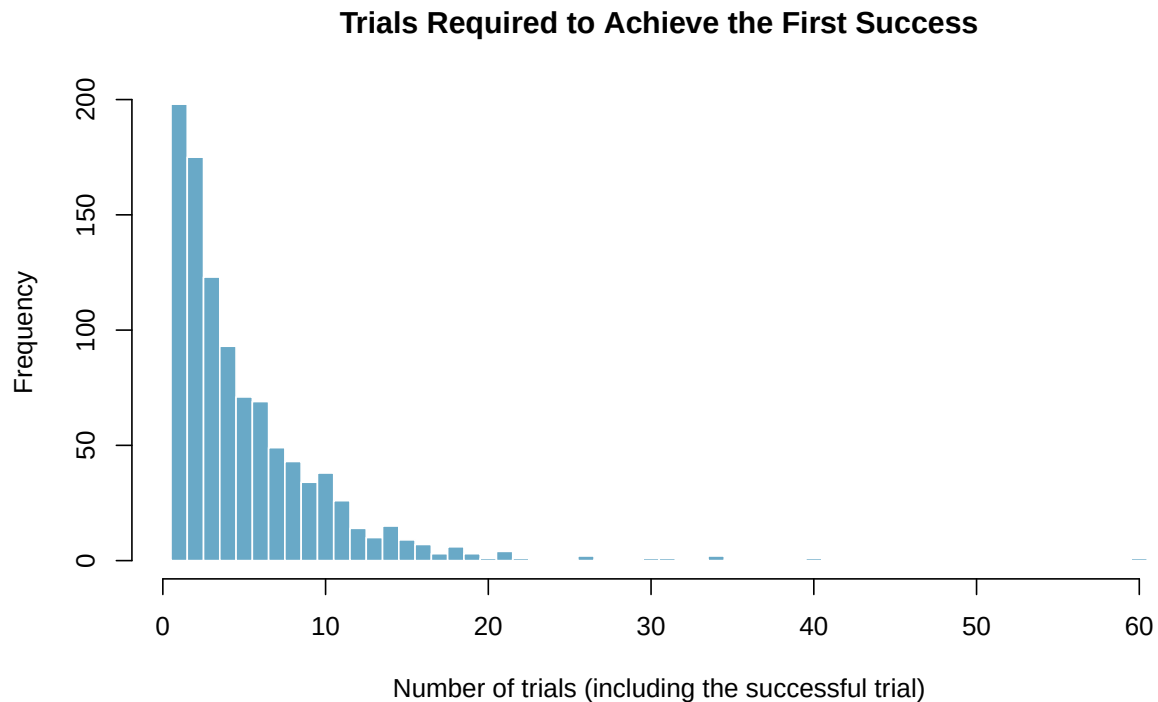
```
## Variance (in 2 decimal places): 24.40
```

```
cat(sprintf("Standard deviation (in 2 decimal places): %.2f", sd_x))
```

```
## Standard deviation (in 2 decimal places): 4.94
```

5. Plot the histogram

```
# Half-integer boundaries put one integer trial count in each bin.
hist(
  x,
  breaks = seq(0.5, max(x) + 0.5, by = 1),
  freq = TRUE,
  col = "#69A9C7",
  border = "white",
  main = "Trials Required to Achieve the First Success",
  xlab = "Number of trials (including the successful trial)",
  ylab = "Frequency"
)
```



The distribution is right-skewed. Short waits are more common while some sequences require many trials before the first success.

II. Hypergeometric Distribution

Consider a plant manufacturing IC chips of which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box. Estimate the probability that the sample contains more than 10% defectives, when:

1. A sample of 10 from a box of 40

Here, $N = 40$, $K = 4$, $N - K = 36$, and $s = 10$. Therefore,

$$P(D > 1) = 1 - \frac{\binom{4}{0}\binom{36}{10} + \binom{4}{1}\binom{36}{9}}{\binom{40}{10}}.$$

```
box_40 <- 40
sample_size <- 10
defective_40 <- 4
good_40 <- box_40 - defective_40

# More than 10% defective in a sample of 10 means more than 1 defective.
prob_40 <- phyper(q = 1, m = defective_40, n = good_40,
                  k = sample_size, lower.tail = FALSE)

# Independently evaluate the complement using the probability mass function.
p0_40 <- dhyper(0, m = defective_40, n = good_40, k = sample_size)
p1_40 <- dhyper(1, m = defective_40, n = good_40, k = sample_size)
check_40 <- 1 - p0_40 - p1_40
stopifnot(isTRUE(all.equal(prob_40, check_40)))

cat(sprintf("P(D = 0), box of 40: %.2f", p0_40))
```

```
## P(D = 0), box of 40: 0.30
```

```
cat(sprintf("P(D = 1), box of 40: %.2f", p1_40))
```

```
## P(D = 1), box of 40: 0.44
```

```
cat(sprintf("P(D > 1), box of 40: %.2f", prob_40))
```

```
## P(D > 1), box of 40: 0.26
```

```
cat(sprintf("Probability as a percentage: %.2f%%", 100 * prob_40))
```

```
## Probability as a percentage: 25.59%
```

The probability is approximately **0.26** or **25.59%**. This is the chance that a random sample of 10 chips contains at least 2 of the box's 4 defective chips.

2. A sample of 10 from a box of 5000

Here, $N = 5000$, $K = 500$, $N - K = 4500$, and $s = 10$. Therefore,

$$P(D > 1) = 1 - \frac{\binom{500}{0}\binom{4500}{10} + \binom{500}{1}\binom{4500}{9}}{\binom{5000}{10}}.$$

```
box_5000 <- 5000
defective_5000 <- 500
good_5000 <- box_5000 - defective_5000

prob_5000 <- phyper(q = 1, m = defective_5000, n = good_5000,
                    k = sample_size, lower.tail = FALSE)

# Check the upper-tail result by calculating 1 - P(D = 0) - P(D = 1).
p0_5000 <- dhyper(0, m = defective_5000, n = good_5000, k = sample_size)
p1_5000 <- dhyper(1, m = defective_5000, n = good_5000, k = sample_size)
check_5000 <- 1 - p0_5000 - p1_5000
stopifnot(isTRUE(all.equal(prob_5000, check_5000)))

cat(sprintf("P(D = 0), box of 5000: %.2f", p0_5000))
```

```
## P(D = 0), box of 5000: 0.35
```

```
cat(sprintf("P(D = 1), box of 5000: %.2f", p1_5000))
```

```
## P(D = 1), box of 5000: 0.39
```

```
cat(sprintf("P(D > 1), box of 5000: %.2f", prob_5000))
```

```
## P(D > 1), box of 5000: 0.26
```

```
cat(sprintf("Probability as a percentage: %.2f%%", 100 * prob_5000))
```

```
## Probability as a percentage: 26.39%
```

The probability is approximately **0.26** or **26.39%**. This is the chance that a random sample of 10 chips contains at least 2 of the box's 500 defective chips.